Tyler Bissonnette

Albany, NY 12204 • (802) 497-7672 • tjbissonnette@outlook.com • subalchemy.org

Electrical and Computer Engineering senior specializing in analog and mixed-signal hardware design, sensor front-ends, and PCB prototyping. Combines research in 4-layer board layout, LTspice circuit simulation, and precision bench characterization with an industry background in high-yield cleanroom semiconductor manufacturing.

EDUCATION

University at Albany, State University of New York (SUNY) / Albany, NY • **Expected: Dec 2026**
B.S. in Electrical & Computer Engineering
Cumulative GPA: **3.4 / 4.0**

TECHNICAL SKILLS

Hardware & Prototyping: Analog Circuit Design, KiCad (2- and 4-Layer PCB Routing), Surface-Mount Soldering (down to 0402), Through-Hole Soldering, Rapid Prototyping, 3D Printing, Hardware-Software Integration

Test & Measurement: Oscilloscopes, Differential Probes, Function Generators, Benchtop Power Supplies, Digital Multimeters (DMM), Circuit Debugging, Cleanroom ESD Protocols

Software & Simulation: SPICE / LTspice, C, C++, Python (NumPy, SoundDevices), MATLAB, OpenCV, SQLite, R Studio, SolidWorks

TECHNICAL & ENGINEERING PROJECTS

Analog Synthesizer Fabrication & Tooling / *Independent Engineering Project* Albany, NY • **2023 – Present**

- **Precision SMD Assembly:** Hand-assembled high-density analog synthesizer modules using 0402 SMD passives, discrete transistors, and op-amps; troubleshooting signal paths using bench oscilloscope.
- **Electro-Mechanical Integration:** Fabricated a cordless vacuum solder stencil fixture from open-source CAD models, engineering custom wiring, power distribution, and flight controller integration without schematics.

Continuous Sound Level Meter (SLM) / *Capstone Design Project* Albany, NY • **2026 – Present**

- **Acoustic Instrumentation:** Engineered an acoustic measurement system adhering to ANSI S1.4 and S1.11 Class-2 specifications, translating acoustic compliance standards into physical circuit and filtering constraints.
- **Sensor Front-End & DSP:** Interfaced an analog microphone and precision ADC with a Raspberry Pi 4B; implemented real-time Python DSP filters (SoundDevices) to compute A- and C-weighted sound levels (dB SPL).

Solid-State Switching & Gate Drive Research / *Undergraduate Research Assistant* Albany, NY • **June 2025 – Present**

- **LTspice Circuit Simulation:** Modeled zero-voltage crossing dynamics in LTspice, calculating optimal LC ratios to mitigate transient voltage spikes during fast fault isolation.
- **PCB Design & Bench Verification:** Designed 2- and 4-layer prototype PCBs in KiCad for active analog gate drivers and a circuit breaker; validated switching response of gate drives using differential probes.

Automated TCG Inventory & Vision Sorter / *Independent Embedded Project* Albany, NY • **2026**

- **Computer Vision Pipeline:** Developed a C++ application with OpenCV and Tesseract OCR to automatically deskew, extract set IDs, and catalog inventory via contour detection and adaptive thresholding.

PROFESSIONAL EXPERIENCE

GlobalFoundries / *CMP Operator & Technical Trainer* Essex Junction, VT • **June 2020 – Sept 2021**

- **Cleanroom Operations:** Operated CMP wafer manufacturing equipment in a Class-100 cleanroom, strictly adhering to hazardous chemical handling, contamination control, and ESD protocols.
- **High-Yield Compliance:** Executed SOPs to hit production yield targets with zero tolerance for wafer defectivity.
- **Technical Leadership:** Promoted to New-Hire Technical Trainer to onboard, instruct, and qualify incoming operators on tooling safety, SOPs, and quality metrics.